2023-2024学年人教版八年级数学上册第13章轴对称测试卷（含答案）

（满分：120分　　时间100分钟）

一、选择题（本大题共10小题，共30分）

1. 下列图案不是轴对称图形的是(　)

A. 　**B.** 　**C.** 　**D.**

2. 等腰三角形的一个内角为$70°$，则另外两个内角的度数分别是(　)

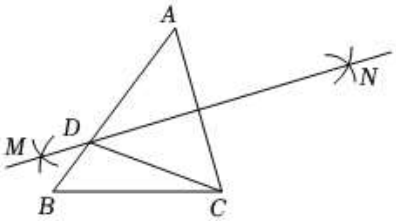
A. $55°$，$55°$　　**B.** $70°$，$40°$或$70°$，$55°$

C. $70°$，$40°$　　**D.** $55°$，$55°$或$70°$，$40°$

3. 在平面直角坐标系中，将点$A(-3,-2)$向右平移5个单位长度得到点B，则点B关于y轴的对称点B'的坐标为(　)

A. $(2,2)$　　**B.** $(-2,2)$　　**C.** $(-2,-2)$　　**D.** $(2,-2)$

4. 如图，在△ABC中，$AB=8$，$BC=6$，以A，C为圆心、大于$\frac{1}{2}AC$的长为半径画圆弧，两弧相交于点M，N，过M，N作直线与AB相交于点D，则△BCD的周长为(　)



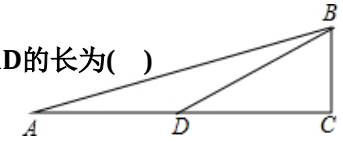
A. 7　　**B.** 10

C. 11　　**D.** 14

5. 若点$A(m-n,m-2n)$与点$B(m-3n,1-\frac{1}{2}m)$关于y轴对称，则点$P(m,n)$所在象限为(　)

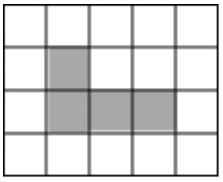
A. 第一象限　　**B.** 第二象限　　**C.** 第三象限　　**D.** 第四象限

6. 如图，在△ABC中，$\angle C=90°$，$\angle A=15°$，$\angle DBC=60°$，$BC=1$，则AD的长为(　)



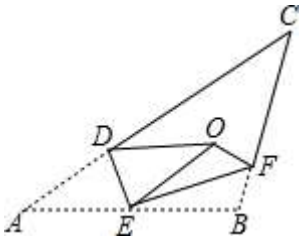
A. 1.5　　**B.** 2　　**C.** 3　　**D.** 4

7. 如图所示是4×5的方格纸，请在其中选取一个白色的方格并涂黑，使图中阴影部分是一个轴对称图形，这样的涂法有(　)



A. 4种　　B. 3种　　C. 2种　　D. 1种

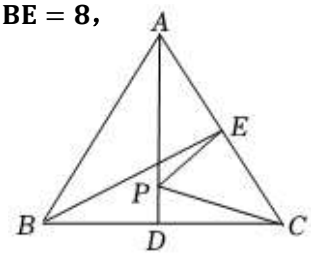
8. 如图，将△ABC沿DE、EF翻折，顶点A，B均落在点O处，且EA与EB重合于线段EO，若∠CDO + ∠CFO = 100°，则∠C的度数为(　)



A. 40°　　B. 41°

C. 42°　　D. 43°

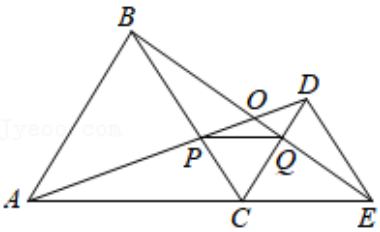
9. 如图所示，△ABC中，AB = AC，AD，BE是△ABC的两条中线，AD = 6，BE = 8，P是AD上的一个动点，连接PE，PC，则CP + EP的最小值是(　)



A. 6　　B. 7

C. 8　　D. 9

10. 如图，已知ΔABC和ΔCDE都是等边三角形，且A、C、E三点共线．AD与BE交于点O，AD与BC交于点P，BE与CD交于点Q，连结PQ.以下六个结论：①AD = BE；②∠AOB = 60°；③AP = BQ；④DE = DP；⑤PQ//AE；⑥OC平分∠AOE.其中正确结论的有个．(　)

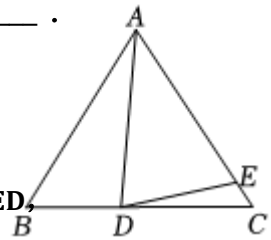


A. 3　　B. 4　　C. 5　　D. 6

二、填空题（本大题共 5 小题，共 15 分）

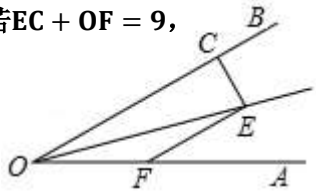
11. 在平面直角坐标系中，若点P(3, a)和点Q(b, −2)关于x轴对称，则a + b的值为_____ .

12. 已知等腰三角形一腰上高与另一腰夹角30°，则顶角的度数为_____.

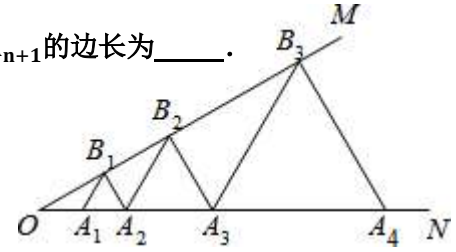


13. 如图，在△ABC中，∠B = ∠C，D为BC边上的一点，点E在AC边上，∠ADE = ∠AED，若∠BAD = 30°，则∠CDE的度数为_____.

14. 如图，∠AOE = ∠BOE = 15°，EF//OB交OA于F点，EC ⊥ OB于C点，若EC + OF = 9，则EF = _____.



15. 如图，$\angle MON=30^{\circ}$，点A_1，A_2，A_3，…在射线ON上，点B_1，B_2，B_3，…在射线OM上，$\triangle A_1B_1A_2$，$\triangle A_2B_2A_3$，$\triangle A_3B_3A_4$ …均为等边三角形．若$OA_1=1$，则$\triangle A_nB_nA_{n+1}$的边长为_____．



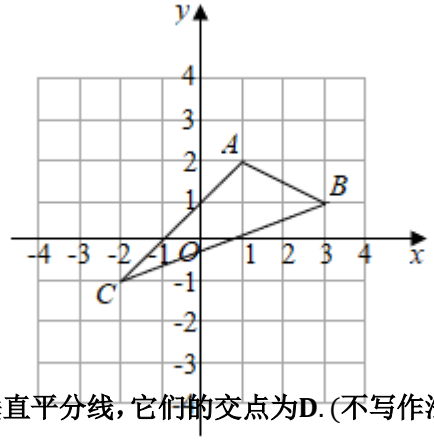
三、解答题（本大题共 8 小题，共 75 分）

16. (8分)如图，在平面直角坐标系中．

(1)作$\triangle ABC$关于x轴对称的$\triangle A_1B_1C_1$；

(2)求出$\triangle ABC$的面积；

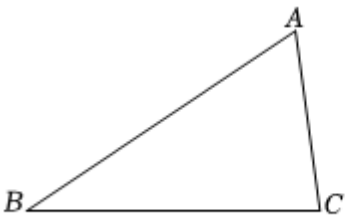
(3)在x轴上是否存在一点P，使得$\triangle AA_1P$与$\triangle ABC$面积相等？若存在，请求出点P的坐标；若不存在，说明理由．

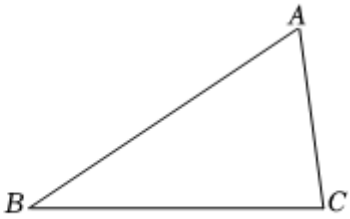


17. (8.分)已知：如图所示$\triangle ABC$．

(1)请在图中用无刻度的直尺和圆规作图：作$\angle BAC$的平分线和BC的垂直平分线，它们的交点为D．(不写作法，保留作图痕迹)

(2)若$AB=15$，$AC=9$，过点D画$DE\perp AB$，则BE的长为_____．(如需画草图，请使用备用图)





（备用图）

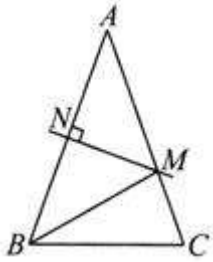
18. (9分)如图，在$\triangle ABC$中，已知$AB=AC$，AB的垂直平分线交AB于点N，交AC于点M，连接MB．

(1)若$\angle ABC=70^{\circ}$，则$\angle NMA$的度数是_____．

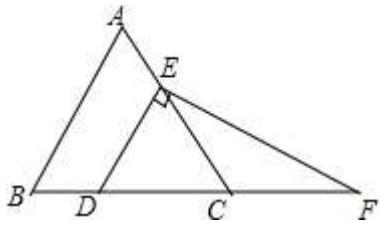
(2)若$AB=8\ cm$，$\triangle MBC$的周长是$14\ cm$．

①求BC的长度；

②若P为直线MN上一点，请你直接写出$\triangle PBC$周长的最小值．



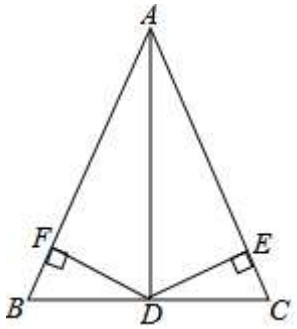
19. (9分)如图，在等边△ ABC中，点D、E分别在边BC、AC上，且DE//AB，过点E作EF ⊥ DE，交BC的延长线于点F.求证：DC = CF.



20. (10 分)如图所示，已知点D为△ ABC的边BC的中点，DE ⊥ AC，DF ⊥ AB，垂足分别为点E，F.且BF = CE.

求证：(1)∠B = ∠C；

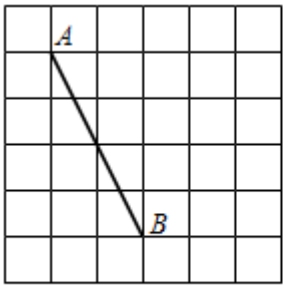
(2)AD平分∠BAC.



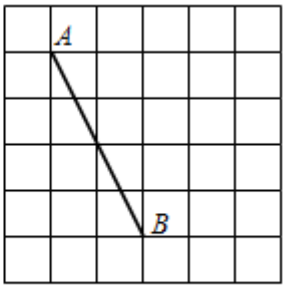
21. (10 分)图①、图②均是6 × 6的正方形网格，小正方形的边长为1，每个小正方形的顶点称为格点，点A、B均在格点上．只用无刻度的直尺，在给定的网格中按要求画图．

(1)在图①中，画一个以AB为底边的等腰三角形ABC，点C在格点上；

(2)在图②中，画一个以AB为腰的等腰三角形ABD，点D在格点上．



图①



图②

22. (10 分) 课本再现：(1)如图1，△ ABC是等边三角形，DE//BC，分别交AB，AC于点D，E.求证：△ ADE是等边三角形．

课本中给出一种证明方法如下： 证明：∵△ ABC是等边三角形， ∴ ∠A = ∠B = ∠C. ∵ DE//BC， ∴ ∠ADE = ∠B，∠AED = ∠C， ∴ ∠A = ∠ADE = ∠AED， ∴△ ADE是等边三角形. “想一想，本题还有其他证法吗？”	给出的另外一种证明方法，请补全： 证明：∵△ ABC是等边三角形， ∴ ∠B = ∠C，∠A = 60°. ∵ DE//BC， ∴ ∠B = ∠ADE，∠C = ① _____ ， ∴ ② _____ = ③ _____ ， ∴ AD = AE. (④_____) ∴△ ADE是等腰三角形. 又∵ ∠A = 60°，∴△ ADE是等边三角形.

(2)如图2，等边三角形ABC的两条角平分线相交于点D，延长BD至点E，使得AE = AD，求证：△ ADE是等边三角形.

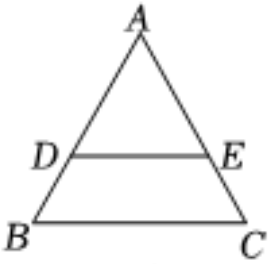


图1

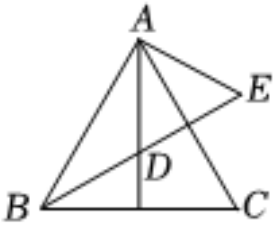
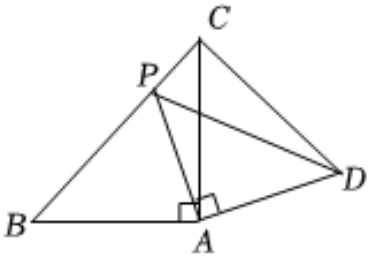
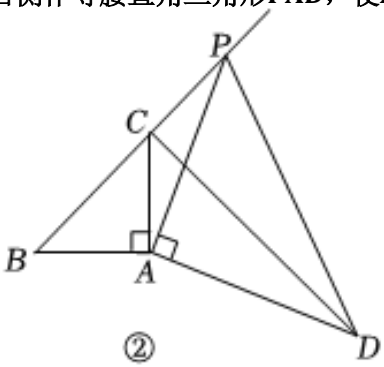


图2

23. (11 分)等腰直角三角形ABC中，AB = AC，∠BAC = 90°，P为射线BC上的一个动点(不与点B，C重合)，连接AP，以AP为直角边，A为直角顶点，在AP右侧作等腰直角三角形PAD，使AP = AD，连接CD.



①



②

(1)如图①，当点P在线段BC上时，求证：△ BAP≌△ CAD；

(2)如图②，当点P在线段BC的延长线上时，请直接写出线段BP和CD的数量关系与位置关系.

参考答案

1.【答案】D

2.【答案】D

解：分情况讨论：

(1)若等腰三角形的顶角为$70°$时，另外两个内角$=(180°-70°)\div 2=55°$；

(2)若等腰三角形的底角为$70°$时，它的另外一个底角为$70°$，顶角为$180°-70°-70°=40°$.

故选：D.

3.【答案】C

解：点$A(-3,-2)$向右平移5个单位长度得到的点B的坐标为$(-3+5,-2)$，即$(2,-2)$，

则点B关于y轴的对称点B'的坐标是：$(-2,-2)$.

故选：C.

4.【答案】D

解：由作法得MN垂直平分AC，

$\therefore DA=DC$，

$\therefore\triangle BCD$的周长$=BD+DC+BC=BD+DA+BC=BA+BC=8+6=14$.

故选：D.

$\triangle BCD$的周长$=BA+BC$.

5.【答案】A

解：$\because$点$A(m-n,m-2n)$与点$B(m-3n,1-\frac{1}{2}m)$关于y轴对称，

$\therefore\begin{cases}m-n+m-3n=0\\m-2n=1-\frac{1}{2}m\end{cases}$，

解得：$\begin{cases}m=2\\n=1\end{cases}$

则点$P(m,n)$所在象限为第一象限.

故选：A.

6.【答案】B

解：$\because\angle DBC=60°$，$\angle C=90°$，

$\therefore\angle BDC=90°-60°=30°$，

$\therefore BD = 2BC = 2 \times 1 = 2$，

$\because \angle C = 90°$，$\angle A = 15°$，

$\therefore \angle ABC = 90° - 15° = 75°$，

$\therefore \angle ABD = \angle ABC - \angle DBC = 75° - 60° = 15°$，

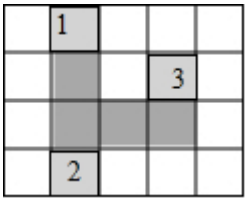
$\therefore \angle ABD = \angle A$，

$\therefore AD = BD = 2$.

故选：B.

7.【答案】B

【解析】解：根据轴对称图形的概念可知，一共有3种涂法，如下图所示：



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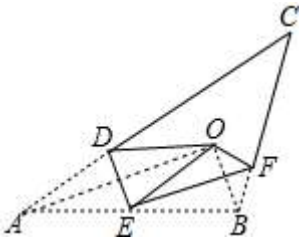
故选：B.

8.【答案】A

解：如图，连接AO、BO.



由题意$EA = EB = EO$，

$\therefore \angle AOB = 90°$，$\angle OAB + \angle OBA = 90°$，

$\because DO = DA$，$FO = FB$，

$\therefore \angle DAO = \angle DOA$，$\angle FOB = \angle FBO$，

$\therefore \angle CDO = 2\angle DAO$，$\angle CFO = 2\angle FBO$，

$\because \angle CDO + \angle CFO = 100°$，

$\therefore 2\angle DAO + 2\angle FBO = 100°$，

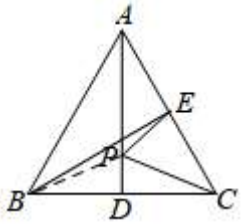
$\therefore \angle DAO + \angle FBO = 50°$，

$\therefore \angle CAB + \angle CBA = \angle DAO + \angle OAB + \angle OBA + \angle FBO = 140°$，

$\therefore \angle C = 180° - (\angle CAB + \angle CBA) = 180° - 140° = 40°$，

故选：A.

9.【答案】C



解：如图，连接PB，

$\because AB = AC$，$BD = CD$，

$\therefore AD \perp BC$，

$\therefore PB = PC$，

$\therefore PC + PE = PB + PE$，

$\because PE + PB \geq BE$，

$\therefore$ P、B、E共线时，$PB + PE$的值最小，最小值为BE的长度，

$CP + EP$的最小值是8.

故选：C.

10.【答案】C

解：① $\because \Delta ABC$ 和 ΔCDE 为等边三角形，

$\therefore AC = BC$， $CD = CE$， $\angle BCA = \angle DCE = 60°$，

$\therefore \angle ACD = \angle BCE$，

$\therefore \Delta ACD \cong \Delta BCE(SAS)$，

$\therefore AD = BE$，

故①正确；

由(1)中的全等得 $\angle CBE = \angle DAC$，

$\because \angle ACB = \angle DCE = 60°$，

$\therefore \angle BCQ = 180° - 60° - 60° = 60° = \angle ACP$，

又 $AC = BC$，

$\therefore \Delta CQB \cong \Delta CPA(ASA)$，

$\therefore AP = BQ$，

故③正确；

$\because \Delta CQB \cong \Delta CPA$，

$\therefore CQ = CP$，

又 $\because \angle PCQ = 60°$，

$\therefore \Delta PCQ$ 为等边三角形，

$\therefore \angle PQC = \angle DCE = 60^\circ$，

$\therefore PQ//AE$，

故⑤正确；

$\because \angle QCP = 60^\circ$， $\angle DPC = \angle BCA + \angle PAC > 60^\circ$，

$\therefore PD \neq CD$，

$\therefore DE \neq DP$，

故④错误；

$\because \angle ACB = \angle CED = 60^\circ$，

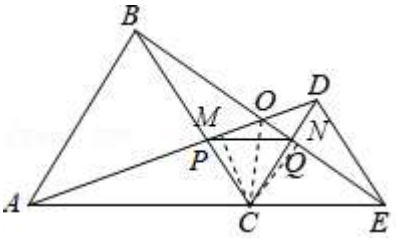
$\therefore BC//DE$，

$\therefore \angle CBE = \angle BED$，

$\because \angle CBE = \angle DAE$，

$\therefore \angle AOB = \angle OAE + \angle AEO = 60^\circ$， 故②正确；

作 $CM \perp AD$， $CN \perp BE$，



$\because \Delta COM \cong \Delta CON$，

$\therefore CM = CN$，

$\therefore OC$ 平分 $\angle AOE$， 故⑥正确，

故正确的有①②③⑤⑥共5个，

故选： C．

11.【答案】5

解：$\because$点$P(3,a)$和点$Q(b,-2)$关于x轴对称，

$\therefore b = 3$，$a = 2$，

则$a + b$的值为：5．

故答案为：5．

12.【答案】120° 或 60°

解：当顶角为钝角时，如图1，可求得其顶角的邻补角为 60°，则顶角为 120°；

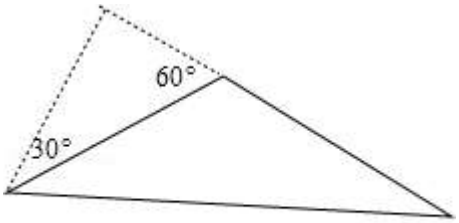


图1

当顶角为锐角时，如图2，可求得其顶角为 60°；

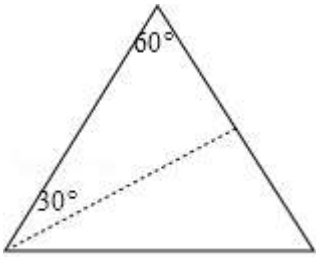


图2

综上可知该等腰三角形的顶角为 120° 或 60°．

故答案为： 120° 或 60°．

13.【答案】15°

解：$\because \angle ADC$是$\triangle ABD$的外角，

$\therefore \angle ADC = \angle B + \angle BAD = \angle B + 30^\circ$，

$\because \angle AED$是$\triangle CDE$的外角，

$\therefore \angle AED = \angle C + \angle CDE$，

$\because \angle B = \angle C$，$\angle ADE = \angle AED$，

$\therefore \angle C + \angle CDE = \angle ADC - \angle CDE = \angle B + 30^\circ - \angle CDE$，

解得$\angle CDE = 15^\circ$．

故答案为：15°．

14.【答案】6

解：作 $EH \perp OA$ 于 H，

$\because \angle AOE = \angle BOE = 15^\circ$， $EC \perp OB$， $EH \perp OA$，

$\therefore EH = EC$， $\angle AOB = 30^\circ$，

$\because EF//OB$，

$\therefore \angle EFH = \angle AOB = 30^\circ$， $\angle FEO = \angle BOE$，

$\therefore EF = 2EH$， $\angle FEO = \angle FOE$，

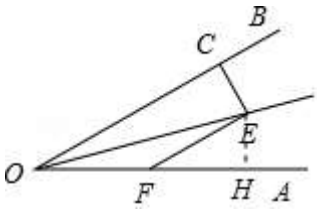
$\because EC + OF = 9$，

$\therefore OF + \frac{1}{2}OF = 9$，

$\therefore OF = 6$，

$\therefore OF = EF = 6$，

故答案为：6．



15.【答案】2^{n-1}

解：$\because \triangle A_1B_1A_2$ 是等边三角形，

$\therefore A_1B_1 = A_2B_1$， $\angle 3 = \angle 4 = \angle 12 = 60^\circ$，

$\therefore \angle 2 = 120^\circ$，

$\because \angle MON = 30^\circ$，

$\therefore \angle 1 = 180^\circ - 120^\circ - 30^\circ = 30^\circ$，

又 $\because \angle 3 = 60^\circ$，

$\therefore \angle 5 = 180^\circ - 60^\circ - 30^\circ = 90^\circ$，

$\because \angle MON = \angle 1 = 30^\circ$，

$\therefore OA_1 = A_1B_1 = 1$，

$\therefore A_2B_1 = 1$，

$\because \triangle A_2B_2A_3$、$\triangle A_3B_3A_4$ 是等边三角形，

$\therefore \angle 11 = \angle 10 = 60^\circ$， $\angle 13 = 60^\circ$，

$\because \angle 4 = \angle 12 = 60^\circ$，

$\therefore A_1B_1 // A_2B_2 // A_3B_3$， $B_1A_2 // B_2A_3$，

$\therefore \angle 1 = \angle 6 = \angle 7 = 30^\circ$， $\angle 5 = \angle 8 = 90^\circ$，

$\therefore A_2B_2 = 2B_1A_2$， $B_3A_3 = 2B_2A_3$，

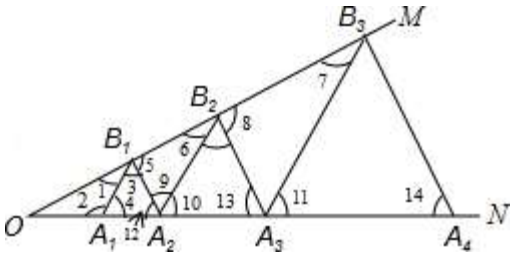
$\therefore A_3B_3 = 4B_1A_2 = 4$，

$A_4B_4 = 8B_1A_2 = 8$，

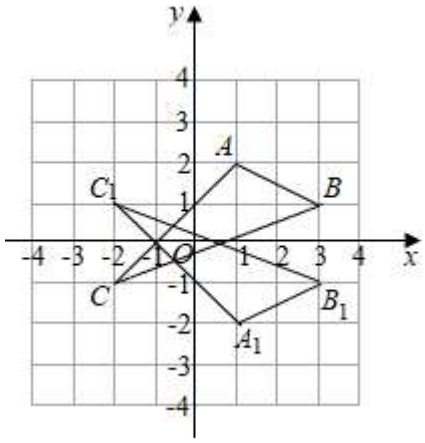
$A_5B_5 = 16B_1A_2 = 16$，

以此类推：$\triangle A_nB_nA_{n+1}$ 的边长为 2^{n-1}．

故答案是：2^{n-1}．



16.【答案】解：(1)如图所示，$\triangle A_1B_1C_1$即为所求；



(2)$S_{\triangle ABC} = \frac{1}{2} \times (1+3) \times 5 - \frac{1}{2} \times 1 \times 2 - \frac{1}{2} \times 3 \times 3 = \frac{9}{2}$;

(3)存在，

设点P坐标为$(a, 0)$，

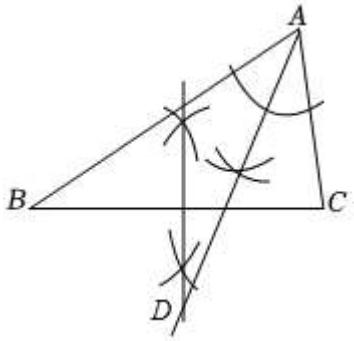
根据题意，得：$\frac{1}{2} \times 4 \times |a-1| = \frac{9}{2}$,

解得$a = \frac{13}{4}$或$a = -\frac{5}{4}$,

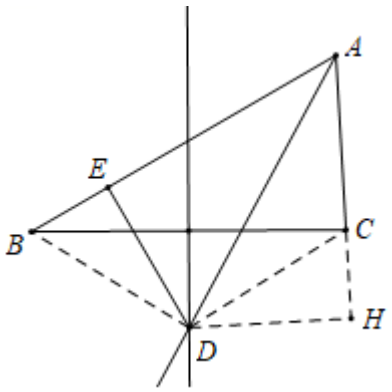
∴点P坐标为$(\frac{13}{4}, 0)$或$(-\frac{5}{4}, 0)$.

17.【答案】3

解：(1)如图，点D即为所求；



(2)如图，连接BD，DC，过点D作DH ⊥ AC交AC的延长线于点H.



∵ AD平分∠ABC，

∴ ∠DAE = ∠DAH，

∵ ∠AED = ∠H = 90°，AD = AD，

∴△ ADE≌△ ADH(AAS)，

∴ AE = AH，DE = DH，

∵点D在线段BC的垂直平分线上，

∴ DB = DC，

∵ ∠DEB = ∠H = 90°，

∴ Rt △ DEB≌Rt △ DHC(HL)，

∴ BE = CH，

∴ AB − AC = AE + BE − (AH − CH) = 2BE = 6，

∴ BE = 3.

故答案为：3.

18..【答案】解：(1)50°；

(2)① ∵ MN是AB的垂直平分线，

$\therefore AM = BM$，

$\therefore \triangle MBC$的周长$= BM + CM + BC = AM + CM + BC = AC + BC$，

$\because AB = 8$，$\triangle MBC$的周长是14，

$\therefore BC = 14 - 8 = 6cm$；

② $\because PB + PC = PA + PC$，$PA + PC \geq AC$，

$\therefore$ P与M重合时，$PA + PC = AC$，此时$PB + PC$最小，

$\therefore \triangle PBC$周长的最小值$= AC + BC = 8 + 6 = 14cm$.

解：(1) $\because AB = AC$，

$\therefore \angle C = \angle ABC = 70°$，

$\therefore \angle A = 40°$，

$\because$ AB的垂直平分线交AB于点N，

$\therefore \angle ANM = 90°$，

$\therefore \angle NMA = 50°$；

(2)①见答案；

②见答案.

19. $\because \triangle ABC$是等边三角形，

$\therefore \angle B = \angle BCA = 60°$，

$\because DE//AB$，

$\therefore \angle EDC = \angle B = \angle A = \angle DEC = 60°$，

$\therefore ED = DC$，

$\because EF \perp DE$，

$\therefore \angle DEF = 90°$，

$\therefore \angle CEF = 90° - \angle DEC = 30°$，

$\therefore \angle F = 90° - \angle EDC = 30°$，

$\therefore EC = CF$，

$\therefore DC = CF$.

20.【答案】证明：

(1) $\because$点D是$\triangle ABC$的边BC的中点，

$\therefore BD = CD$，

$\because DE \perp AC$，$DF \perp AB$，

$\therefore \angle BFD = \angle CED = 90°$，

在$Rt\triangle BDF$和$Rt\triangle CDE$中，

$\begin{cases} BD = CD \\ BF = CE \end{cases}$，

$\therefore Rt\triangle BDF \cong Rt\triangle CDE(HL)$，

$\therefore \angle B = \angle C$.

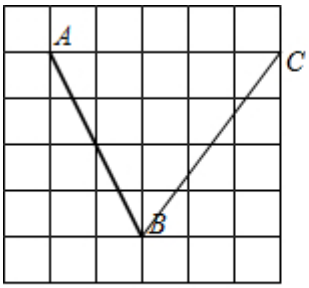
(2) $\because \angle B = \angle C$，

$\therefore AB = AC$，

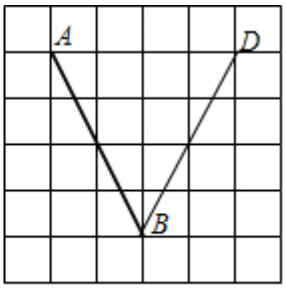
$\because AB = AC$，$BD = DC$，

$\therefore$ AD平分$\angle BAC$.

21.【答案】解：(1)如图①中，$\triangle ABC$即为所求作(答案不唯一).



图①



图②

(2)如图②中，$\triangle ABD$即为所求作(答案不唯一).

(1)作腰为5，底为AB的等腰三角形即可(答案不唯一，顶点C在线段AB的垂直平分线上即可).

(2)根据等腰三角形的定义画出图形即可(答案不唯一).

本题考查作图–应用与设计，等腰三角形的判定等知识，解题的关键是理解题意，灵活运用所学知识解决问题.

22.【答案】解：(1)①$\angle AED$；②$\angle ADE$；③$\angle AED$；④等角对等边；

(2)证明：$\because \triangle ABC$是等边三角形，

$\therefore \angle BAC = \angle ABC = 60°$，

$\because$ BE和AD分别为$\angle ABC$和$\angle BAC$的平分线，

$\therefore \angle ABD = \frac{1}{2}\angle ABC = 30°$，$\angle BAD = \frac{1}{2}\angle BAC = 30°$.

$\because \angle ADE$为$\triangle ABD$的外角，

$\therefore \angle ADE = \angle ABD + \angle BAD = 60°$，

$\because AE = AD$，

$\therefore \triangle ADE$是等边三角形.

解：(1) $\because \triangle ABC$是等边三角形，

$\therefore \angle B = \angle C$，$\angle A = 60°$.

$\because DE//BC$，

$\therefore \angle B = \angle ADE$，$\angle C = \angle AED$，

$\therefore \angle ADE = \angle AED$，

$\therefore AD = AE$(等角对等边)，

$\therefore \triangle ADE$是等腰三角形.

又$\because \angle A = 60°$，

$\therefore \triangle ADE$是等边三角形.

故答案为：①$\angle AED$；②$\angle ADE$；③$\angle AED$；④等角对等边；

23.【答案】解：(1) $\because$在等腰直角三角形BAC与等腰直角三角形PAD中，$\angle PAD = \angle CAB = 90°$，

$\therefore \angle PAC + \angle CAD = \angle PAC + \angle PAB$，

$\therefore \angle CAD = \angle BAP$，

在$\triangle CAD$与$\triangle BAP$中，

$$\begin{cases} CA = BA \\ \angle CAD = \angle BAP \\ AD = AP \end{cases}，$$

$\therefore \triangle CAD\{\cong \triangle BAP(SAS)$.

(2) $\because$在等腰直角三角形BAC与等腰直角三角形PAD中，$\angle PAD = \angle CAB = 90°$，

$\therefore \angle PAC + \angle PAD = \angle PAC + \angle CAB$，

$\therefore \angle CAD = \angle BAP$，

在$\triangle CAD$与$\triangle BAP$中，

$$\begin{cases} CA = BA \\ \angle CAD = \angle BAP \\ AD = AP \end{cases}$$

$\therefore \triangle CAD \cong \triangle BAP(SAS)$，

$\therefore BP = CD$，$\angle PBA = \angle DCA$，

$\because \angle PBA + \angle BCA = 90°$，

$\therefore \angle DCA + \angle BCA = 90°$，即$\angle BCD = 90°$，

$\therefore BP \perp CD$，

综上：$BP = CD$；$BP \perp CD$.